Admissibility and Tail Behaviour in Contrastive Forecasting

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Abstract

Contrastive learning can be read as a methodology for learning a conditional distribution: a learned score tilts the marginal distribution of one modality, and the tilted measure is the model. Adopting this population-level viewpoint, developed by Baptista, Stuart and Tran, we ask which design choices in a contrastive forecaster survive in the limit of infinite data and flexible encoders: the loss (one-way, two-way, or joint), the statistical divergence that it minimizes, or the score. We first show that every loss variant, under every divergence, has the same population-level target, the log density ratio up to a conditioning-dependent constant; combined with the universal approximation theorems of Stuart and Wolf, the loss and the divergence drop out for flexible encoders. We then show that the score does not drop out, and that it matters precisely through its tail behaviour. A score that is bounded above defines a valid model for every data distribution, and the forecast obtained by reweighting a finite bank of observed futures converges to that model as the bank grows. A score that grows without bound in the future, such as the unnormalized inner product, admits no valid model on heavy-tailed data, even with ReLU encoders, and its bank forecast collapses onto a single future as the bank grows. This identifies the tail condition in existing non-compact universal approximation results as necessary rather than technical. Numerical experiments on synthetic and real time series confirm both halves and add one refinement: in practice a ceiling on the score is necessary but not sufficient, and the score must also not reward extreme futures.

Keywords: Contrastive Learning, CLIP, Probabilistic Forecasting, Importance Sampling, Heavy Tails, Model Misspecification

1 Introduction

Contrastive learning trains two encoders, one for each of two paired data modalities, so that matched pairs receive a high score and mismatched pairs a low one. In the forecasting setting considered here, the two modalities are the past u and the future v of a time series. A recent line of work [1, 2] views such algorithms probabilistically: the learned score $s(u, v)$ defines a probability model for v given u, obtained by reweighting the marginal distribution of futures by e^s, and the contrastive losses used in practice are statistical divergences between this model and the data. This paper studies which design choices in the resulting forecaster matter, and why.

In Subsection 1.1 we provide background and review related work. Subsection 1.2 states our contributions and describes the organization of the paper. Subsection 1.3 fixes notation.

1.1 Background and Literature Review

The probabilistic framework. Our starting point is the framework of Baptista, Stuart and Tran [1]. There, a score defines a tilting of the product of the marginals, the CLIP

loss and its one-way and joint variants are identified as Kullback–Leibler (KL) divergences between the data and the tilted model [1, Thm. 3.2, Thm. 3.8, Rem. 3.12], generalizations to other divergences are formulated [1, Opt. Probs. 3.4 and 3.6, Rem. 3.5], and a forecast is obtained by reweighting a finite bank of observed futures [1, Eq. (31)]. For Gaussian data and linear encoders every minimizer is available in closed form, and the choice of loss and of score visibly changes the resulting forecast [1, Sec. 5].

Expressivity. Stuart and Wolf [2] adopt the same population-level view and study expressivity. They show that the joint KL divergence between the data and the model bounds the conditional losses from above [2, Thms. 2.5, 2.6], that uniform approximation of the log density ratio lifts to approximation of the measure in KL [2, Thms. 3.2, 3.3], and that two-tower CLIP, with unnormalized or normalized encoders, is a universal approximator in KL on compact domains [2, Thms. 4.1, 4.2]. Their extension to non-compact domains truncates the approximating score outside a compact set; the underlying estimate requires the approximant to be bounded by a constant B outside that set and carries an error term of order e^B [2, Prop. A.3, Thm. 3.3]. They work with KL throughout and list other divergences, such as those built from the energy score, as future work.

The optimal score. That the optimal InfoNCE critic is the pointwise mutual information plus a function of the conditioning variable is standard [4, 5]. For the symmetric CLIP loss, Uesaka et al. [6] show that the optimal similarity is the pointwise mutual information and argue that inner products of finite-dimensional embeddings may fail to reach it. This is the closest relative of the misspecification view taken in Section 3, but it concerns a single loss rather than agreement or disagreement between variants.

Reweighting candidates, and its failure. Sordoni et al. [7] note that resampling candidates with softmax-normalized critic scores approximates the conditional distribution, exactly as the pool of candidates grows; this is the bank forecast studied here. Weight collapse of self-normalized importance sampling is classical: in high dimension it is studied by Bengtsson, Bickel and Li [12], heavy-tailed weights are diagnosed by Pareto-smoothed importance sampling [13], and the required sample size is of order $e^{D_{\mathrm{KL}}}$ by Chatterjee and Diaconis [14]. That an exponential tilt $e^{\langle a,v\rangle}$ requires a finite moment generating function is textbook. Our results in Section 4 apply these facts to contrastive scores; we claim no new probability.

Bounded critics and normalization. In mutual-information estimation, Song and Ermon [8] clip the critic to tame the variance of partition-function estimates. In representation learning, normalized embeddings are standard because they remove the scale degree of freedom that lets the softmax become arbitrarily sharp [9, 10], and negative squared Euclidean distance logits are used in practice and match or exceed cosine CLIP [11]. To our knowledge none of these works ties boundedness to the *existence* of the model, or to the behaviour of a reweighted bank.

Analog forecasting. Forecasting by weighting historical futures is the analog ensemble of weather forecasting [15] and kernel analog forecasting [16]. A contrastive bank forecast is an analog forecast whose kernel, $e^{s(u,v)}$, is learned.

1.2 Contribution and Paper Organization

The primary contributions of this paper are a population-level account of which design choices in a contrastive forecaster matter, together with numerical evidence on synthetic and real time

series. In detail, we make the following contributions, structured to simultaneously describe the paper organization:

(C1) *Loss and divergence invariance* (Section 3). We show that the one-way, two-way and joint losses, under any divergence, vanish exactly when the score equals the log density ratio up to, respectively, a function of the past, a function of the future, or a constant. Consequently, all of them agree at the population level whenever the score family is well-specified. Combining this with the results of Stuart and Wolf, every KL-based loss variant is simultaneously driven to zero by flexible two-tower encoders on compact domains.

(C2) *Loss differences as misspecification* (Section 3.2). In the Gaussian, linear setting of [1] we show that each score family omits a different ingredient of the log density ratio, which explains the predictions of [1] as artefacts of misspecification. We prove that linear L^2 scores cannot reach zero two-way or joint loss on any correlated Gaussian, and give the scalar optimum in closed form.

(C3) *Admissibility is governed by the tails of the score* (Section 4.1). A score bounded above yields a valid model for every data distribution. A score linear in the future yields no valid model on heavy-tailed data unless it ignores the past, and ReLU encoders do not remove this obstruction. Boundedness has a price: a bounded score can be exact only when the density ratio is bounded.

(C4) *Convergence and collapse of the bank forecast* (Section 4.2). We show that the bank forecast converges as the bank grows if and only if the score is admissible, that it collapses onto a single future when the scores over the bank are regularly varying, and that a two-sided bound guarantees a uniform effective sample size at the cost of capped sharpness. This shows that the tail condition in [2, Prop. A.3] is necessary rather than a convenience of the proof.

(C5) *Numerical experiments* (Section 5). On synthetic Gaussian, heavy-tailed, warped and bimodal processes, and on real weather data, we confirm the loss invariance for flexible encoders, the failure of unbounded scores on heavy tails, and the predicted turnaround of the unbounded forecast as the bank grows. The experiments add one refinement: a ceiling on the score is necessary but not sufficient, and the score must also not reward extreme futures.

Section 6 concludes and lists limitations and open questions. Proofs are collected in Appendix A.

1.3 Notation

For a measurable space we write $\mathcal{P}(\cdot)$ for the set of probability measures on it. The past u takes values in $\mathcal{U} \subseteq \mathbb{R}^{d_u}$, the future v in $\mathcal{V} \subseteq \mathbb{R}^{d_v}$, and $\mathcal{X} := \mathcal{U} \times \mathcal{V}$. The data distribution is $\mu \in \mathcal{P}(\mathcal{X})$, with marginals μ_u and μ_v and product of marginals $\mu_{\mathsf{P}} := \mu_u \otimes \mu_v$. Conditionals are written $\mu_{v|u}(\cdot|u)$ and $\mu_{u|v}(\cdot|v)$. Euclidean inner products and norms are $\langle \cdot, \cdot \rangle$ and $|\cdot|$. For symmetric matrices, $A \preceq B$ denotes the Loewner order. δ_x is the Dirac measure at x, and $\mathbf{1}_A$ the indicator of a set A.

2 Probabilistic Formulation

In this section we specify the data model (Section 2.1), the model defined by a score (Section 2.2), the loss functions (Section 2.3), and the finite-sample forecast that is used in practice

(Section 2.4). The presentation follows [1, 2].

2.1 Problem Formulation

Assumption 2.1 (Data Distribution)**.** The data generating measure $\mu \in \mathcal{P}(\mathcal{X})$ satisfies the following.

(A1) *Marginal distributions.* The marginals $\mu_u \in \mathcal{P}(\mathcal{U})$ and $\mu_v \in \mathcal{P}(\mathcal{V})$ are absolutely continuous with respect to Lebesgue measure.

(A2) *Joint distribution.* The measures μ and μ_{P} are mutually absolutely continuous, with Radon–Nikodym derivative

$$r(u,v) \coloneqq \frac{\mathrm{d}\mu}{\mathrm{d}\mu_{\mathsf{P}}}(u,v) = \frac{p(v|u)}{p(v)}. \tag{1}$$

(A3) *Finite mutual information.* $\log r \in L^1(\mathcal{X},\mu)$, so that $I(u;v) \coloneqq \mathbb{E}_\mu \log r < \infty$.

The function $\log r$ is the pointwise mutual information: it measures how much more likely the future v is after observing the past u than it is a priori. It is the tilting function of [2], and it is the object that every loss studied below aims to recover.

2.2 The Model Defined by a Score

Definition 2.2 (Score and Tilted Model)**.** A *score* is a measurable function $s : \mathcal{X} \to \mathbb{R}$. It defines the joint and conditional measures

$$\nu^s(\mathrm{d}u,\mathrm{d}v) \coloneqq \frac{e^{s(u,v)}}{Z}\,\mu_{\mathsf{P}}(\mathrm{d}u,\mathrm{d}v), \qquad \nu^s_{v|u}(\mathrm{d}v|u) \coloneqq \frac{e^{s(u,v)}}{Z(u)}\,\mu_v(\mathrm{d}v), \tag{2}$$

with normalizing constants

$$Z \coloneqq \int_{\mathcal{X}} e^s \,\mathrm{d}\mu_{\mathsf{P}}, \qquad Z(u) \coloneqq \int_{\mathcal{V}} e^{s(u,v)}\,\mu_v(\mathrm{d}v), \qquad Z'(v) \coloneqq \int_{\mathcal{U}} e^{s(u,v)}\,\mu_u(\mathrm{d}u), \tag{3}$$

and $\nu^s_{u|v}$ defined analogously with $Z'(v)$ [1, Eqs. (13)–(16)]. When $Z < \infty$, the conditional of ν^s given u is $\nu^s_{v|u}$.

In words, the forecast for a past u takes the distribution of all futures and reweights each one by e^{score}. The construction requires the normalizing constants to be finite, which motivates the following definition; it is central to Section 4.

Definition 2.3 (Admissible Score)**.** A score s is *admissible for the forecasting direction* if $Z(u) < \infty$ for μ_u-almost every u, and *admissible* if, in addition, $Z'(v) < \infty$ for μ_v-almost every v and $Z < \infty$. If a normalizing constant is infinite, the corresponding model does not exist, and we assign the value $+\infty$ to every loss that involves it.

Example 2.4 (Scores)**.** Let $f : \mathcal{U} \to \mathbb{R}^{d_z}$ and $g : \mathcal{V} \to \mathbb{R}^{d_z}$ be encoders of the past and the future, and let $\kappa > 0$ be a learnable scale (the inverse temperature, $\kappa = 1/\tau$ in [2]). We consider

$$s_{\mathrm{dot}} = \kappa\,\langle f(u),\, g(v)\rangle\,, \qquad s_{\mathrm{ncos}} = \kappa\left\langle \frac{f(u)}{|f(u)|},\, \frac{g(v)}{|g(v)|}\right\rangle, \qquad s_{L^2} = -\frac{\kappa}{2}\,|f(u)-g(v)|^2\,. \tag{4}$$

The unnormalized inner product s_{dot} is the score of [2, Thm. 4.1], the normalized cosine s_{ncos} is that of CLIP [3] and of [2, Thm. 4.2], and s_{L^2} is the L^2 score of [1]. Note that $s_{L^2} \le 0$ and $|s_{\mathrm{ncos}}| \le \kappa$, whereas s_{dot} is unbounded whenever g is.

2.3 Population-Level Loss Functions

Definition 2.5 (Divergence)**.** A *divergence* is a map $D : \mathcal{P} \times \mathcal{P} \to [0, \infty]$ with $D(p, q) = 0$ if and only if $p = q$.

The KL divergence is a divergence. So is the energy distance, on distributions with a finite first moment,

$$D_{\mathrm{E}}(p, q) \coloneqq 2\,\mathbb{E}\,|X - Y| - \mathbb{E}\,|X - X'| - \mathbb{E}\,|Y - Y'|\,, \qquad X, X' \sim p,\ Y, Y' \sim q \text{ independent}, \tag{5}$$

as is the maximum mean discrepancy with a characteristic kernel. The energy distance is also our evaluation metric: the expected energy score [17] of a forecast q, minus that of the truth p, equals $\frac{1}{2} D_{\mathrm{E}}(p, q)$.

Definition 2.6 (Contrastive Losses)**.** For a divergence D, weights $\lambda_u, \lambda_v > 0$ and a score s, define the *one-way*, *reverse one-way*, *two-way* and *joint* losses

$$\begin{aligned} J^D_{v|u}(s) &\coloneqq \mathbb{E}_{u \sim \mu_u}\, D\big(\mu_{v|u}(\cdot|u), \nu^s_{v|u}(\cdot|u)\big), & J^D_{u|v}(s) &\coloneqq \mathbb{E}_{v \sim \mu_v}\, D\big(\mu_{u|v}(\cdot|v), \nu^s_{u|v}(\cdot|v)\big), \\ J^D_2(s) &\coloneqq \lambda_u J^D_{v|u}(s) + \lambda_v J^D_{u|v}(s), & J^D_{\mathrm{joint}}(s) &\coloneqq D\big(\mu, \nu^s\big). \end{aligned} \tag{6}$$

When $D = D_{\mathrm{KL}}$ we drop the superscript.

Remark 2.7 (Relation to CLIP and InfoNCE). With $D = D_{\mathrm{KL}}$, expanding (6) with (2) gives $J_{v|u}(s) = I(u; v) - \mathbb{E}_\mu[s] + \mathbb{E}_{\mu_u}[\log Z(u)]$. Replacing the measures by empirical measures yields the InfoNCE loss for $J_{v|u}$, the CLIP loss for J_2 with $\lambda_u = \lambda_v = \frac{1}{2}$, and a single softmax over all pairs for J_{joint} [1, Thm. 3.2, Thm. 3.8, Rem. 3.12]; see also [2, Sec. 2.2]. For general D the losses correspond to [1, Opt. Probs. 3.4 and 3.6]; they can be trained from samples alone when D is the KL divergence, an energy distance, or a maximum mean discrepancy [1, Rem. 3.5].

2.4 The Bank Forecast

The model (2) involves the full marginal μ_v. In practice it is replaced by a finite collection of observed futures.

Data Assumption 2.8. We have access to a *bank* $v_1, \ldots, v_N \overset{\text{iid}}{\sim} \mu_v$ of futures, independent of the data used to train the score.

Following [1, Eq. (31)], the *bank forecast* for a past u is

$$\nu^N_{v|u}(\cdot|u) \coloneqq \sum_{j=1}^{N} w_j\, \delta_{v_j}, \qquad w_j \coloneqq \frac{e^{s(u, v_j)}}{\sum_{k=1}^{N} e^{s(u, v_k)}}, \qquad \mathrm{ESS} \coloneqq \Big(\sum_{j=1}^{N} w_j^2\Big)^{-1}. \tag{7}$$

This is self-normalized importance sampling with proposal μ_v and target $\nu^s_{v|u}$. The effective sample size $\mathrm{ESS} \in [1, N]$ counts how many bank futures the forecast effectively uses; $\mathrm{ESS} \approx 1$ means that the forecast has collapsed onto a single future.

3 Loss and Divergence Invariance

In this section we show that every loss variant, under every divergence, aims at the same object, the log density ratio. The variants differ only in which part of it they leave free, and the divergences differ only in how they measure a miss. Differences between them are therefore a symptom of a score family that cannot represent $\log r$.

Proposition 3.1 (Common Target)**.** *Let μ satisfy Assumption 2.1, let D be any divergence, and let s be an admissible score. Then*

(i) $J^D_{v|u}(s) = 0$ *if and only if* $s(u,v) = \log r(u,v) + c(u)$ *for some function* c*;*

(ii) $J^D_{u|v}(s) = 0$ *if and only if* $s(u,v) = \log r(u,v) + c'(v)$ *for some function* c'*;*

(iii) $J^D_2(s) = 0$ *if and only if* $J^D_{\text{joint}}(s) = 0$ *if and only if* $s = \log r + \text{const}$*.*

All identities hold μ_{P}-almost everywhere. The right-hand sides do not involve D.

Proof. See Appendix A.1. □

Definition 3.2 (Well-Specified Family)**.** A family $\mathcal{S}$ of scores is *well-specified* for a loss if that loss attains the value 0 on $\mathcal{S}$. By Proposition 3.1, this means that $\mathcal{S}$ contains some $\log r + c(u)$ for the one-way loss, and $\log r + \text{const}$ for the two-way and joint losses, whatever the divergence.

Corollary 3.3 (Agreement)**.** *If $\mathcal{S}$ contains $\log r + \text{const}$, then for every divergence D the one-way, two-way (for any $\lambda_u, \lambda_v > 0$) and joint losses all have minimizers whose forecast is exactly the true conditional, $\nu^s_{v|u} = \mu_{v|u}$. Neither the loss variant nor the divergence can be distinguished at the population level.*

The converse fails: Section 3.2 contains a family that cannot represent $\log r$, yet for which the one-way and two-way losses agree.

3.1 Flexible Encoders

Corollary 3.3 requires the family to contain $\log r$ exactly. For flexible encoders, the results of Stuart and Wolf provide the approximate analogue for the KL divergence.

Theorem 3.4 (Approximate Agreement, KL)**.** *Let μ satisfy Assumption 2.1 with $\mathcal{U}$ and $\mathcal{V}$ compact, and let the encoders be drawn from a parametric family satisfying the C^0-universal approximation assumption of [2, Sec. 3]. Then for every $\epsilon > 0$ there exist a latent dimension d_z and encoders f, g such that $s = s_{\text{dot}}$ satisfies*

$$J_{\text{joint}}(s) < \epsilon, \qquad J_{v|u}(s) < \epsilon, \qquad J_{u|v}(s) < \epsilon, \qquad J_2(s) < (\lambda_u + \lambda_v)\,\epsilon.$$

The same holds for $s = s_{\text{ncos}}$, with κ chosen large enough.

Proof. See Appendix A.2. □

A single flexible score therefore makes every KL-based loss variant small simultaneously. This is the first half of our account: once the encoders are flexible, the loss variant drops out. The statement is restricted to compact domains, on which continuous encoders produce bounded scores; Section 4 studies what happens when that boundedness is lost.

Theorem 3.4 does not transfer to other divergences, because closeness in KL does not imply closeness in other divergences. The simplest case is a forecast that is correct except for a small amount of mass moved far away.

Proposition 3.5 (KL Is Blind to How Far Misplaced Mass Goes)**.** *Let p be a distribution on $\mathbb{R}^d$ with a finite mean and no atom at $a \in \mathbb{R}^d$, and let $q_\varepsilon = (1-\varepsilon)p + \varepsilon\delta_a$. Then*

$$D_{\text{KL}}(p \,\|\, q_\varepsilon) \le -\log(1-\varepsilon) \quad \text{for every } a,$$

whereas

$$\tfrac{1}{2} D_{\text{E}}(p, q_\varepsilon) = \tfrac{1}{2}\varepsilon^2\big(2\,\mathbb{E}\,|X-a| - \mathbb{E}\,|X-X'|\big) \xrightarrow[|a|\to\infty]{} \infty.$$

Proof. See Appendix A.3. □

Moving a fraction ε of the forecast to an arbitrarily extreme future costs at most about ε in KL, while its cost in energy score grows linearly with the distance. A model trained with KL therefore has almost no incentive to keep mass out of the far tails, whereas an evaluation by energy score penalizes exactly that. The divergences thus agree exactly at the optimum (Corollary 3.3) and approximately in the bulk (Theorem 3.4), but an approximation that is good in one may be poor in another in the tails. Training with the energy distance itself, as [1, Rem. 3.5] allows, would penalize tail over-weighting directly.

3.2 Rigid Families: the Gaussian Case

When the family is not flexible, the loss variants come apart. The Gaussian, linear setting of [1, Sec. 5] makes this explicit. There, every minimizer is available in closed form, and four predictions are relevant here:

(P1) the inner-product score gets the mean of the forecast right but its spread wrong, which remains at the unconditional level [1, Cor. 5.2];

(P2) the one-way L^2 score gets both mean and spread right [1, Thm. 5.3, Cor. 5.4];

(P3) for the inner-product score, the one-way and two-way losses give the same answer [1, Thm. 5.1];

(P4) the joint loss shrinks the forecast towards the mean and is a worse forecaster than the one-way loss [1, Thm. 5.6, Cor. 5.7].

We now show that each prediction is a consequence of misspecification.

Let μ be a centred Gaussian with covariance blocks C_{uu}, C_{uv}, C_{vv}, joint covariance C, joint precision $P = C^{-1}$, and $C_0 \coloneqq \operatorname{blockdiag}(C_{uu}, C_{vv})$, the covariance of μ_{P}. Then

$$\log r(u,v) = -\tfrac{1}{2}u^\top (P_{uu} - C_{uu}^{-1})u \;-\; u^\top P_{uv} v \;-\; \tfrac{1}{2}v^\top (P_{vv} - C_{vv}^{-1})v + \text{const.} \tag{8}$$

There are three ingredients: a past–future interaction, which sets the mean of the forecast; a future-only term, which sets its spread; and a past-only term. With linear encoders every score is of the form $s = -\frac{1}{2}u^\top Ru + u^\top Av - \frac{1}{2}v^\top Sv$, and the families differ in which ingredients they can set (Table 1).

family	allowed (R, A, S)	well-specified for
inner product	$R = S = 0$	nothing (if $C_{uv} \neq 0$)
L^2	$\begin{pmatrix} R & -A \\ -A^\top & S \end{pmatrix} = [G, -H]^\top [G, -H] \succeq 0$	one-way loss only
inner product + quadratics	R, S symmetric, A free	every loss

Table 1: Linear score families on Gaussian data. For L^2, $f(u) = Gu$ and $g(v) = Hv$, with κ absorbed.

Table 1 recovers the predictions. For P1: the inner product has no future-only term, so the covariance of its forecast is C_{vv} whatever A is. For P4: the joint loss also needs the past-only term, which the inner product lacks; moreover, the joint normalizer is finite only if $\left\| C_{uu}^{1/2} A C_{vv}^{1/2} \right\| < 1$, which is the origin of the shrinkage function of [1]. For P2: the one-way L^2 family is well-specified, because $S \succeq 0$ can match the future-only term. The two-way and joint L^2 losses are not.

Proposition 3.6 (Linear L^2 Is Too Rigid for the Two-Way and Joint Losses)**.** *Under linear L^2 scores, every model joint covariance satisfies $(C_0^{-1}+Q)^{-1} \preceq C_0$, with $Q = [G, -H]^\top [G, -H] \succeq 0$. The data satisfy $C \not\preceq C_0$ whenever $C_{uv} \neq 0$. Hence the two-way and joint losses cannot reach zero.*

Proof. See Appendix A.4. □

In words, an L^2 model can couple past and future only by shrinking both marginals, while real data couple them without shrinking.

Proposition 3.7 (Scalar Closed Form)**.** *For unit-variance scalars with correlation $\rho \in (0, 1)$, the joint-L^2 optimum forecasts $\mathcal{N}\big(\frac{\rho}{2-\rho}u, \frac{2(1-\rho)}{2-\rho}\big)$, against the truth $\mathcal{N}(\rho u, 1-\rho^2)$. The mean is shrunk and the forecast is over-confident. The fraction of the mutual information $I(u; v)$ that it loses tends to $3/8$ as $\rho \to 0$, and to 0 as $\rho \to 1$.*

Proof. See Appendix A.5. □

		inner product			L^2			+ quad.
setting	$I(u; v)$	1-way	2-way	joint	1-way	2-way	joint	all three
$\rho = 0.1$	0.005	0.005	0.005	0.005	0	0.294	0.333	0
$\rho = 0.3$	0.047	0.046	0.046	0.051	0	0.219	0.251	0
$\rho = 0.5$	0.144	0.131	0.131	0.157	0	0.148	0.170	0
$\rho = 0.7$	0.337	0.272	0.272	0.323	0	0.081	0.092	0
$\rho = 0.9$	0.830	0.512	0.512	0.571	0	0.021	0.023	0
GP of Section 5	1.627	0.704	0.704	0.746	0	0.003	0.003	0

Table 2: Fraction of the mutual information $I(u; v)$ (in nats) lost by the forecast at the population optimum of each KL loss: 0 is perfect, 1 is the unconditional forecast. Values are exact, computed from Gaussian KL divergences without training or sampling. The last row uses the exact covariances of the synthetic Gaussian process of Section 5.

Table 2 makes the picture quantitative, and two points stand out. First, each rigid family fails in its own regime. The error of the inner product lies in the spread, which matters most when dependence is strong; the error of two-way and joint L^2 lies in the coupling-by-shrinking constraint, which matters most when dependence is weak. Second, adding the missing ingredients removes every difference: giving the inner product free past-only and future-only quadratic terms makes all three losses exact (last column). This is the flexibility argument of Section 3.1 in miniature.

4 Admissibility and the Tails of the Score

Flexibility concerns representing $\log r$ where the data are. It says nothing about how the score behaves far from the data, and that is where the choice of score continues to matter. The non-compact universal approximation result of [2, Thm. 3.3] rests on an estimate [2, Prop. A.3] which requires the approximating score to satisfy $|s| \leq B$ outside a compact set $\mathcal{K}$, and whose error contains a term of order e^B; the theorem then enforces the bound by truncating the approximant to zero outside $\mathcal{K}$. For a two-tower score and $\mathcal{K} = \mathcal{K}_u \times \mathcal{K}_v$, truncation corresponds to encoders that vanish outside compact sets, since $\mathbf{1}_{\mathcal{K}} \langle f, g \rangle = \langle \mathbf{1}_{\mathcal{K}_u} f, \mathbf{1}_{\mathcal{K}_v} g \rangle$. Standard neural encoders do not vanish. In this section we show that the tail condition is not a convenience of the proof: without it the model need not exist, and the bank forecast collapses.

4.1 Admissibility

Proposition 4.1 (Bounded Above Implies Admissible)**.** *If $s \le c$, then $Z(u) \le e^c$, $Z'(v) \le e^c$ and $Z \le e^c$ for every data distribution μ. In particular, s_{L^2} (with $c = 0$) and s_{ncos} (with $c = \kappa$) are admissible for every μ, with any encoders.*

Proof. See Appendix A.6. □

Proposition 4.2 (A Score Linear in the Future Needs Light Tails)**.** *Let $s(u, v) = \langle a(u),\, Hv\rangle + b(u)$. Then $Z(u) = e^{b(u)} M_{\mu_v}(H^\top a(u))$, where M_{μ_v} is the moment generating function of μ_v. If μ_v is a multivariate Student-t distribution, with any number of degrees of freedom, then $M_{\mu_v}(w) = \infty$ for every $w \neq 0$. Hence the linear inner-product score is admissible for the forecasting direction only if $H^\top a(u) = 0$ for μ_u-almost every u, that is, only if it ignores the past.*

Proof. See Appendix A.7. □

Proposition 4.3 (ReLU Encoders Do Not Restore Admissibility)**.** *Let $s = s_{\mathrm{dot}}$ with g a ReLU network, and let g_∞ denote the same network with its biases removed. If $\langle f(u),\, g_\infty(v_0)\rangle > 0$ for some direction v_0, and μ_v is an elliptical Student-t distribution, then $Z(u) = \infty$.*

Proof. See Appendix A.8. □

The inner-product score is therefore admissible at u only if it does not grow in any direction of the future, and nothing in training enforces this. A bounded output layer on g, such as tanh, makes the score admissible by Proposition 4.1 without normalizing anything; Section 5.4 shows that this is not sufficient in practice.

Boundedness is not free. Combining Proposition 3.1 with an upper bound on the score constrains the density ratios that the family can represent exactly.

Proposition 4.4 (Bounded Scores Represent Only Bounded Ratios)**.** *Let $s \le c$ be admissible and let D be any divergence.*

(i) If $J^D_{v|u}(s) = 0$, then for μ_u-almost every u, $r(u, \cdot) \le e^c / Z(u)$ holds μ_v-almost everywhere.

(ii) If $J^D_{\mathrm{joint}}(s) = 0$, then $r \le e^c / Z$ holds μ_{P}-almost everywhere.

(iii) If $|s| \le \kappa$ and $J^D_{\mathrm{joint}}(s) = 0$, then $e^{-2\kappa} \le r \le e^{2\kappa}$ holds μ_{P}-almost everywhere.

Proof. See Appendix A.9. □

For a correlated Gaussian, r is unbounded on $\mathcal{X}$, so no score bounded above, however flexible, is well-specified for the joint or two-way loss; Proposition 3.6 is a special case. This does not contradict Theorem 3.4: the approximants constructed in [2, Thm. 3.3] are themselves bounded, being continuous on $\mathcal{K}$ and zero outside it. Boundedness thus costs exactness but not approximability in KL. For the normalized cosine, part (iii) is the counterpart of the observation in [2, Thm. 4.2] that universality requires the temperature $1/\kappa$ to tend to zero.

4.2 Convergence and Collapse of the Bank Forecast

Admissibility is not a technicality: it is exactly the condition under which the bank forecast (7) works.

Theorem 4.5 (Admissibility Is Equivalent to Bank Convergence)**.** *Fix u and let the bank satisfy Data Assumption 2.8. If $Z(u) < \infty$, then $\nu^N_{v|u} \to \nu^s_{v|u}$ weakly, almost surely, as $N \to \infty$, and $\mathrm{ESS}/N \to 1/\mathbb{E}_{\mu_v}\left[(\mathrm{d}\nu^s_{v|u}/\mathrm{d}\mu_v)^2\right]$ when this moment is finite. If $Z(u) = \infty$, then the empirical normalizer $\frac{1}{N}\sum_j e^{s(u,v_j)}$ diverges and there is no limit to converge to.*

Proof. See Appendix A.10. □

Proposition 4.6 (Collapse)**.** *If the scores $X_j = s(u, v_j)$ of the bank futures have a regularly varying right tail, $\mathbb{P}(X > x) = x^{-\alpha}L(x)$ with L slowly varying, then $\max_j w_j \to 1$ in probability as $N \to \infty$. This covers the linear inner-product score on Student-t futures, with α equal to the degrees of freedom.*

Proof. See Appendix A.11. □

Proposition 4.7 (Bounded on Both Sides: Guaranteed Bank, Capped Sharpness)**.** *If $|s| \le \kappa$, then $e^{-2\kappa} \le \mathrm{d}\nu^s_{v|u}/\mathrm{d}\mu_v \le e^{2\kappa}$. Hence $D_{\mathrm{KL}}\left(\nu^s_{v|u} \,\|\, \mu_v\right) \le 2\kappa$ and, in the limit $N \to \infty$, $\mathrm{ESS}/N \ge e^{-2\kappa}$, uniformly in u. The price is that no forecast can be more than $e^{2\kappa}$ times as concentrated as the marginal μ_v, or carry more than 2κ nats of information.*

Proof. See Appendix A.12. □

The two outcomes are therefore precise. With a bounded score the bank forecast converges to the model as the bank grows. With an unbounded score on heavy-tailed data it collapses onto the single most extreme future, and a larger bank only supplies more extreme futures to collapse onto. Table 3 checks this for fixed scores, with no training, and Table 4 summarizes the hierarchy.

bank size N	linear, Gaussian	linear, Student-t ($\nu = 3$)	L^2, Student-t ($\nu = 3$)
10^2	0.081 (0.41)	0.240 (0.30)	0.018 (0.79)
10^3	0.016 (0.38)	0.455 (0.038)	0.002 (0.79)
10^4	0.003 (0.37)	0.760 (0.0007)	$< 10^{-3}$ (0.79)
10^5	0.001 (0.37)	0.902 (10^{-5})	$< 10^{-4}$ (0.79)
10^6	$< 10^{-4}$ (0.37)	0.954 (10^{-6})	$< 10^{-5}$ (0.79)

Table 3: Largest bank weight (mean over 200 banks), with ESS/N in brackets, for a fixed score $s = v$ or $s = -\frac{1}{2}(v-1)^2$ on unit-variance futures. For the Gaussian, Theorem 4.5 gives $\mathrm{ESS}/N \to e^{-1} \approx 0.368$.

	unbounded (s_{dot})	bounded above (s_{L^2})	bounded on both sides (s_{ncos})
model exists	only with light-tailed futures	always	always
bank forecast converges	only with light-tailed futures	always	always
bank size needed	can be infinite (collapse)	finite, $\approx e^{D_{\mathrm{KL}}(\nu_{v\|u} \,\|\, \mu_v)}$ [14]	uniformly $\lesssim e^{2\kappa}$
sharpness	unlimited	unlimited	capped by κ

Table 4: The boundedness hierarchy.

Remark 4.8 (Why Training Does Not Avoid Inadmissibility). At the population level an inadmissible parameter has infinite loss, so an exact minimizer would avoid it. Training, however, uses the empirical loss, which is always finite: a batch of 256 futures almost never contains the extreme futures that make $Z(u)$ infinite. Training can therefore settle on an informative parameter whose score grows in the future, and the problem appears only when the forecast meets a large bank. For the linear inner product on Student-t data, Proposition 4.2 shows that every informative parameter is of this kind. Proposition 3.5 gives a second reason: even when the model exists, KL training barely charges for misplaced tail mass.

5 Numerical Experiments

We test the predictions of Sections 3 and 4 on synthetic processes, each of which relaxes one assumption of the Gaussian setting, and on real weather data. Section 5.1 describes the setup. Section 5.2 tests the loss invariance of Section 3, Sections 5.3 and 5.4 test the tail results of Section 4, and Section 5.5 discusses finite data.

5.1 Setup

Evaluation. Every contrastive model is evaluated only through its own score, by building the bank forecast (7) from a bank of real futures not used in training. Forecasts are scored with the energy score [17], a proper scoring rule for multivariate probabilistic forecasts that rewards both accuracy and calibrated uncertainty. On synthetic data each domain has an exact optimal forecaster (oracle), and we report the energy score as a percentage above that of the oracle (0% is perfect). On real data we report *skill*, the fractional reduction in energy score against a reference forecast (higher is better).

Synthetic domains. Past and future are windows of 50 points of a smooth Gaussian process observed with noise. Each synthetic domain has a parameter whose zero point is this Gaussian process, the setting of [1]:

- *heavy tails*: an elliptical Student-t process with ν degrees of freedom ($\nu = \infty$ is the Gaussian process; $\nu = 3$ is the heaviest used);
- *bent GP*: a monotone warp of the Gaussian process, giving skewed futures and a non-linear relation between past and future;
- *double-well*: a stochastic process that jumps between two levels, so that the future can have two separate likely outcomes.

Each synthetic setting uses 200k training pairs, a bank of 50k held-out futures, 500 test cases and 2 seeds, so that differences below about 1% are noise.

Real data. We use one year (2020) of 10-minute readings from the Jena weather station. We forecast the next 6 h of temperature from the last day of six variable groups, and the next 6 h of rain from past pressure, wind and moisture.

Encoders. Linear encoders reproduce the setting of [1]. Flexible encoders are two-layer MLPs of width 256 with GELU activations and a linear shortcut.

5.2 Loss Variants

Linear encoders: the loss and the score both matter. On the Gaussian process, the inner-product score is +5.6% above perfect whichever direction it is trained in (P3), and +9.8% when trained jointly (P4). Its 90% bands contain the truth 98% of the time on the hardest third of cases, i.e. they are too wide (P1). The L^2 score is at +0.0% in all three variants, consistent with the last row of Table 2. The linear normalized cosine is poor everywhere (+12–23%), because it sees the past only through its direction and loses its level.

Flexible encoders: the loss variant mostly stops mattering. With MLP encoders the range between the variants collapses (Table 5), as Theorem 3.4 predicts, and P1 fades: the coverage of the inner product on the hardest cases drops from 98% to 89%. The best models (joint L^2, normalized cosine) are within about 1% of perfect on every domain. The differences that remain are small and consistently favour the joint loss; we return to them in Section 5.5.

domain	encoder	inner prod.	1-way	2-way	joint	range
Gaussian process	linear		+5.6	+5.6	+9.8	4.2
	MLP		+2.7	+1.4	+0.3	2.4
bent GP (full warp)	linear		+37.4	+40.0	+25.2	14.8
	MLP		+5.9	+3.9	+0.6	5.3
double-well ($\lambda = 1.5$)	linear		+1.3	+2.6	+2.7	1.4
	MLP		+0.6	+0.3	+0.5	0.3

Table 5: Energy score above the oracle (%) for the inner-product score trained with each loss variant. The range across variants shrinks with flexible encoders.

5.3 Unbounded Scores on Heavy Tails

Every catastrophic failure we observed is an unbounded score on heavy tails (Table 6). The linear inner product places essentially all its weight on single bank futures (minimum ESS of 1.0). MLP encoders do not rescue it, in line with Proposition 4.3: MLP inner products are still +7 to +23% above perfect and still collapse onto single futures for some inputs. On rain, which is mostly zero with occasional bursts, the unnormalized inner product scored 1.4–2.4 times worse than climatology, while L^2 and normalized cosine retained the shape "mostly dry, sometimes a burst". Normalization helps exactly where tails are heavy, and only there.

heavy tails, $\nu = 3$	inner product, 2-way	normalized cosine	joint L^2
linear	$\sim 10\times$ worse than oracle	+23.0	+4.5
MLP	+23.3	+0.9	+0.7

Table 6: Energy score above the oracle (%) on the heavy-tailed domain.

5.4 Bank-Size Sweep

The two experiments in this subsection test Theorem 4.5 and Proposition 4.6 directly. Each model is trained as in Section 5.1 (15k steps, batch size 256) and then frozen; only the bank changes. The forecast is rebuilt from held-out banks of size $N = 100$ up to $N = 200{,}000$, with up to 3 disjoint banks per size, on the same 500 test pasts. We record the energy score

above the oracle, the largest bank weight and, for collapsed forecasts (largest weight > 0.5), how extreme the winning future is, measured by its percentile by size within the bank. We use 2 seeds on the heavy-tailed domain ($\nu = 3$) and 1 seed on the Gaussian control.

Does a larger bank help or hurt? Theorem 4.5 and Proposition 4.6 predict that bounded scores improve and level off as the bank grows, and that the unbounded inner product turns around and gets worse. The prediction holds (Table 7). On heavy tails, the unbounded inner product is best at a small bank and then gets steadily worse: with MLP encoders from +7% at 3k futures to +46% at 200k, and with linear encoders from +33% to over +3000%. Every bounded score improves and levels off. On the Gaussian control the same unbounded score also improves and levels off (at +5.7% for linear encoders, matching Section 5.2), so the turnaround is caused by the tails and not by the score alone. The collapse diagnostic shows the mechanism directly: with linear encoders, 20% of the forecasts of the inner product at $N = 200$k put more than half their weight on one future, and that future is the most extreme in the bank (99.998th percentile by size, seed 1).

Is a ceiling enough? Proposition 4.1 implies that a tanh output layer on the future encoder makes the inner product admissible without normalizing anything. The result is only a partial fix (Table 7, second row of each block). It is far better than the uncapped inner product (+103% against +3358% with linear encoders, +7.5% against +46% with MLPs, at $N =$ 200k), but it still turns around: in one MLP seed it improves to +1.2%, in the other it rises from +8% to +14%, and with linear encoders it rises from +22% to +103%. Its forecasts rarely collapse onto one future, but the futures they favour are still the most extreme in the bank (99.99th percentile, linear, seed 1).

This does not contradict Section 4. The ceiling makes the model exist, so the bank forecast converges; but it converges to a model that over-weights extremes. A tanh score rises *towards* its ceiling as the future becomes more extreme, so extreme futures in the favoured direction receive the highest score. A small bank contains few of them and hides the problem; a large bank exposes it.

	model	$N = 100$	1k	3k	30k	100k	200k
heavy tails, MLP	inner product (unbounded)	+33.6	+8.7	**+7.1**	+17.5	+34.5	+46.3
	inner product, tanh future	+37.0	+9.7	+6.7	**+3.2**	+5.4	+7.5
	L^2 one-way	+36.8	+11.8	+8.0	+4.5	+3.8	+2.9
	L^2 joint	+31.5	+6.2	+3.2	+0.9	+0.7	+0.6
	normalized cosine	+41.8	+8.0	+3.5	+1.0	+0.8	+0.8
heavy tails, linear	inner product (unbounded)	**+33.2**	+38.7	+70.2	+1253	+2705	+3358
	inner product, tanh future	+28.0	**+22.1**	+24.5	+64.1	+92.3	+102.6
	L^2 one-way	+22.8	+8.1	+5.8	+4.6	+4.3	+4.2
	L^2 joint	+21.5	+7.8	+5.7	+4.5	+4.5	+4.5
	normalized cosine	+25.4	+23.5	+23.2	+23.1	+23.0	+23.1
Gaussian, MLP	inner product (unbounded)	+13.7	+2.9	+1.5	+2.0	+1.8	+1.8
Gaussian, linear	inner product (unbounded)	+9.9	+6.4	+6.1	+5.8	+5.7	+5.7

Table 7: Energy score above the oracle (%) as the bank grows, for frozen models. Mean of 2 seeds (1 seed for the Gaussian control). Bold marks the best bank size where a curve turns around.

What the two experiments show together. The scores that work either *penalize* extreme futures (s_{L^2} decreases as $g(v)$ moves away from $f(u)$) or are *indifferent* to how extreme

a future is (s_{ncos} depends only on the direction of $g(v)$). It is not the height of the ceiling that matters: the learned scale of the normalized cosine is about $\kappa \approx 72$, so the guarantee of Proposition 4.7 is vacuous, and yet it is the best model here. The underlying reason is Remark 4.8: training cannot correct the tails. A batch of 256 futures almost never contains one from the far tail, so how the score treats extreme futures is determined by the architecture, not learned from data. A ceiling is necessary, since without one the model does not exist; in practice the score must also not *reward* extremity. We have not stated or proved this refinement as a theorem.

5.5 Finite Data: the Loss Matters Again

The results of Section 3 are statements about infinite data. With finite data the loss variant can matter again, through over-fitting rather than through what it aims at. On weather data, MLPs over-fit within a few hundred to a few thousand steps. There, the joint L^2 loss clearly beats the one-way L^2 loss (Table 8), although one-way L^2 is the well-specified variant.

skill (2 seeds)	L^2 1-way	L^2 2-way	L^2 joint
rain, linear (vs climatology)	-0.106	$+0.011$	$+0.012$
temperature, MLP (vs persistence)	$+0.152$	$+0.243$	$+0.280$
temperature, linear (vs persistence)	$+0.352$	$+0.344$	$+0.383$

Table 8: Skill of the L^2 score on Jena weather data (higher is better).

The population analysis bounds what joint and two-way L^2 can lose to rigidity: in the scalar Gaussian, at most about 3/8 (joint) or 1/3 (two-way) of $I(u;v)$ (Proposition 3.7). On rain, $I(u;v)$ is tiny (the best model beats climatology by 1%), so that cost is negligible, and the observed gaps must come from finite data. One-way L^2 has nothing that prevents it from becoming over-sharp, and on rain its forecasts collapsed (minimum ESS of 1.6 with linear encoders, median ESS of 3 with MLPs). Two-way and joint L^2 do have such a check. By a Schur complement, $S \succeq A^\top R^+ A$, so a sharp forecast must come with a sharp backward conditional, which the two-way loss penalizes; the joint loss also scores the model marginals, which a collapsed forecast distorts. On this reading, the small bias of the joint loss buys a regularizer against over-sharpening. This predicts that the deficit of the one-way loss shrinks as the training data grow, as it does on the plentiful synthetic data of Section 5.2. It is a hypothesis, not a result.

6 Conclusion and Future Work

We have viewed a contrastive forecaster as a learned tilting of the marginal distribution of futures, and asked which of its design choices survive in the population limit. The loss and the divergence do not: every variant aims at the log density ratio, and for flexible encoders on compact domains a single score drives every KL-based variant to zero. The score does survive, and precisely through its tails. A score bounded above defines a valid model for every data distribution, and its bank forecast converges; a score that grows in the future defines no model on heavy-tailed data, even with ReLU encoders, and its bank forecast collapses onto the most extreme future. This identifies the tail bound in the non-compact universal approximation result of [2] as necessary rather than technical. Experiments on synthetic and real time series confirm both halves, and suggest that a bounded score must, in addition, not reward extreme futures.

Limitations. Sections 3 and 4 are population-level statements (infinite data, infinitely many negatives); the finite-sample statements are Theorem 4.5 and Proposition 4.6, the experiments of Section 5.4, and the hypothesis of Section 5.5. The information losses in Table 2 are KL divergences, while the experiments use the energy score, which is less sensitive to errors in spread; the two should agree in sign but not in size. Proposition 4.3 is proved for ReLU networks, while our MLPs use GELU, which grows like ReLU at infinity; we expect the same conclusion but have not written out the proof. Rain is physically bounded, so its true $Z(u)$ is finite and Proposition 4.6 does not apply literally; what applies is its finite-N version, a small ESS/N whenever the scores over the bank are heavy-tailed. All experiments use 2 seeds, except the Gaussian control of Section 5.4 (1 seed), and the collapse diagnostic was recorded for seed 1 only.

Open questions. First, the divergence part of Section 3 is proved but not tested: every model here was trained with KL. The natural test is to train with the energy-distance loss permitted by [1, Rem. 3.5]; by Proposition 3.5 we expect this to matter most for scores that reward extreme futures. Second, the "does not reward extremity" refinement of Section 5.4 is an empirical reading of two experiments on one domain and should be stated and proved as a theorem. Third, it remains to establish non-compact universal approximation for an architecture whose score is bounded by construction, such as the L^2 or normalized-cosine two-tower families, rather than by truncation. Finally, the weak-dependence prediction of Proposition 3.7 at finite data, the quadratic-augmented inner product (Table 2, last column) in training, and the bank sweep on rain remain untested.

A Proofs

A.1 Proof of Proposition 3.1

Because $D(p, q) \geq 0$ with equality only if $p = q$, each loss vanishes exactly when the distributions that it compares are equal. The remainder of the argument is therefore about equality of distributions and does not involve D.

(i) $J^D_{v|u}(s) = 0$ if and only if $\nu^s_{v|u}(\cdot|u) = \mu_{v|u}(\cdot|u)$ for μ_u-almost every u. Comparing densities with respect to μ_v, this reads $e^{s(u,\cdot)}/Z(u) = r(u, \cdot)$, that is, $s = \log r + \log Z(u)$. Conversely, $s = \log r + c(u)$ gives $Z(u) = e^{c(u)} \int r(u, v)\, \mu_v(\mathrm{d}v) = e^{c(u)}$, and hence the true conditional. (ii) is symmetric. (iii) Since $\lambda_u, \lambda_v > 0$, $J^D_2 = 0$ if and only if (i) and (ii) both hold, so that $c(u) = c'(v)$ for μ_{P}-almost every (u, v). A function of u alone that equals a function of v alone almost everywhere under a product measure is almost everywhere constant. For the joint loss, $D(\mu, \nu^s) = 0$ if and only if $\nu^s = \mu$, if and only if $e^s/Z = r$, that is, $s = \log r + \log Z$. □

A.2 Proof of Theorem 3.4

By [2, Thm. 4.1] there exist d_z and continuous encoders f, g such that $s = \langle f, g \rangle$ (with $\kappa = 1$ absorbed into the encoders) satisfies $D_{\mathrm{KL}}(\mu \,\|\, \nu^s) < \epsilon$. Since $\mathcal{U}, \mathcal{V}$ are compact and s is continuous, s is bounded, so s is admissible and $\mu \ll \nu^s$. The measure ν^s has density e^s/Z with respect to μ_{P}, so its u-marginal has density $Z(u)/Z$ with respect to μ_u, and its conditional given u is $\nu^s_{v|u}$ of (2); likewise for v. Applying [2, Thm. 2.6] with $m = 2$ to the pairs (v, u) and (u, v) gives $J_{v|u}(s) \leq D_{\mathrm{KL}}(\mu \,\|\, \nu^s)$ and $J_{u|v}(s) \leq D_{\mathrm{KL}}(\mu \,\|\, \nu^s)$, and the bound on J_2 follows. For s_{ncos}, use [2, Thm. 4.2] in place of [2, Thm. 4.1], with $\kappa = 1/\tau$. □

A.3 Proof of Proposition 3.5

Since p has no atom at a, $q_\varepsilon \geq (1-\varepsilon)p$ on the support of p, so $\mathrm{d}p/\mathrm{d}q_\varepsilon \leq 1/(1-\varepsilon)$ p-almost everywhere and $D_{\mathrm{KL}}(p \,\|\, q_\varepsilon) = \int \log(\mathrm{d}p/\mathrm{d}q_\varepsilon)\,\mathrm{d}p \leq -\log(1-\varepsilon)$. For the energy distance, write $m = \mathbb{E}\,|X - X'|$ and $A = \mathbb{E}\,|X - a|$. With $Y \sim q_\varepsilon$, $\mathbb{E}\,|X - Y| = (1-\varepsilon)m + \varepsilon A$ and $\mathbb{E}\,|Y - Y'| = (1-\varepsilon)^2 m + 2\varepsilon(1-\varepsilon)A$. Substituting, $D_{\mathrm{E}} = 2(1-\varepsilon)m + 2\varepsilon A - m - (1-\varepsilon)^2 m - 2\varepsilon(1-\varepsilon)A = \varepsilon^2(2A - m)$, and $A \geq |a| - \mathbb{E}\,|X| \to \infty$. □

A.4 Proof of Proposition 3.6

The model joint has density proportional to $e^s\,\mathcal{N}(0, C_0)$ with $s = -\frac{1}{2}z^\top Q z$ and $z = (u, v)$, that is, it is $\mathcal{N}\big(0, (C_0^{-1} + Q)^{-1}\big)$. Since $C_0^{-1} + Q \succeq C_0^{-1}$, inverting reverses the order, so $(C_0^{-1} + Q)^{-1} \preceq C_0$. On the data side, $C_0 - C = \begin{pmatrix} 0 & -C_{uv} \\ -C_{uv}^\top & 0 \end{pmatrix}$ has eigenvalues $\pm\sigma_i(C_{uv})$, so $C \preceq C_0$ forces $C_{uv} = 0$. Zero two-way or joint loss requires the model joint to equal μ (Proposition 3.1(iii)), which is impossible when $C_{uv} \neq 0$. The inequality $S \succeq A^\top R^+ A$ used in Section 5.5 is the generalized Schur complement criterion for $\begin{pmatrix} R & -A \\ -A^\top & S \end{pmatrix} \succeq 0$. □

A.5 Proof of Proposition 3.7

With $s = -\frac{1}{2}(gu - hv)^2$ and unit variances, $Q = \begin{pmatrix} g^2 & -gh \\ -gh & h^2 \end{pmatrix}$ and, up to constants, $2J_{\text{joint}} = g^2 + h^2 - 2gh\rho - \log(1 + g^2 + h^2)$. For fixed $t = g^2 + h^2$ the cross term is minimized at $g = h$, leaving $t(1-\rho) - \log(1+t)$, which is minimized at $t = \rho/(1-\rho)$. The forecast then has precision $1 + t/2 = (2-\rho)/(2(1-\rho))$ and mean coefficient $\rho/(2-\rho)$. Expanding for small ρ gives $J_{v|u} = \frac{3}{16}\rho^2 + O(\rho^3)$ against $I(u; v) = \frac{1}{2}\rho^2 + O(\rho^4)$, so the ratio tends to $3/8$. At $\rho = 0.5$ the closed form gives 0.1699, matching Table 2. □

A.6 Proof of Proposition 4.1

$Z(u) = \int e^{s(u,v)}\,\mu_v(\mathrm{d}v) \leq e^c$, and likewise for $Z'(v)$ and Z. For L^2, $-\frac{\kappa}{2}\,|f - g|^2 \leq 0$; for the normalized cosine, $\big|\langle \bar{f}, \bar{g} \rangle\big| \leq 1$ by the Cauchy–Schwarz inequality. □

A.7 Proof of Proposition 4.2

$Z(u) = e^{b(u)}\,\mathbb{E}\exp\langle H^\top a(u), v\rangle$ by definition. A multivariate t distribution is elliptical, so for $w \neq 0$ the projection $\langle w, v\rangle$ is a scaled univariate t variable, whose density decays polynomially; hence $\mathbb{E}e^{\langle w, v\rangle} = \infty$. □

A.8 Proof of Proposition 4.3

Write $g = W_L\sigma(\cdots\sigma(W_1 v + b_1)\cdots) + b_L$ with $\sigma = \mathrm{ReLU}$. The ReLU is 1-Lipschitz and positively homogeneous, so by induction over the layers $|g(v) - g_\infty(v)| \leq \sum_\ell \|b_\ell\| \prod_{k>\ell} \|W_k\| =: B < \infty$, and $g_\infty(tv) = t\,g_\infty(v)$ for $t \geq 0$. Let $\delta = \langle f(u), g_\infty(v_0/\,|v_0|)\rangle > 0$. By continuity there is an open cone $\mathcal{K} \ni v_0$ on which $\langle f(u), g_\infty(v/\,|v|)\rangle \geq \delta' > 0$, so on $\mathcal{K}$, $s(u, v) \geq \kappa\big(\delta'\,|v| - |f(u)|\,B\big)$. An elliptical t distribution gives every open cone a polynomial tail, $\mu_v(\mathcal{K} \cap \{|v| > t\}) \asymp t^{-\nu}$, so $\int_{\mathcal{K}} e^{\kappa\delta'|v|}\,\mathrm{d}\mu_v = \infty$. □

A.9 Proof of Proposition 4.4

(i) By Proposition 3.1(i) and its proof, $r(u, \cdot) = e^{s(u,\cdot)}/Z(u) \leq e^c/Z(u)$. (ii) By Proposition 3.1(iii), $r = e^s/Z \leq e^c/Z$. (iii) If $|s| \leq \kappa$ then $Z \in [e^{-\kappa}, e^\kappa]$, so $r = e^s/Z \in [e^{-2\kappa}, e^{2\kappa}]$. □

A.10 Proof of Theorem 4.5

For bounded continuous φ, the strong law of large numbers gives $\frac{1}{N}\sum_j e^{s(u,v_j)}\varphi(v_j) \to \int e^s \varphi \, \mathrm{d}\mu_v$ and $\frac{1}{N}\sum_j e^{s(u,v_j)} \to Z(u)$ almost surely, since both integrands are integrable when $Z(u) < \infty$. The ratio converges to $\int \varphi \, \mathrm{d}\nu^s_{v|u}$, and a countable convergence-determining class gives weak convergence. With $\rho = e^s/Z(u)$, $N\sum_j w_j^2 = \frac{\frac{1}{N}\sum_j \rho(v_j)^2}{(\frac{1}{N}\sum_j \rho(v_j))^2} \to \mathbb{E}_{\mu_v}[\rho^2]$. If $Z(u) = \infty$, the strong law for nonnegative random variables gives $\frac{1}{N}\sum_j e^{s(u,v_j)} \to \infty$. □

A.11 Proof of Proposition 4.6

Let $X_{(1)} \geq X_{(2)}$ be the two largest scores and b_N the $1 - 1/N$ quantile of X, so that $b_N = N^{1/\alpha}\ell(N)$ with ℓ slowly varying. By the extreme-value limit for regularly varying tails, $(X_{(1)}, X_{(2)})/b_N$ converges in distribution to $(\Gamma_1^{-1/\alpha}, \Gamma_2^{-1/\alpha})$, the first two points of a Poisson process, which are almost surely distinct. So for every $\varepsilon > 0$ there is $c > 0$ with $\mathbb{P}(X_{(1)} - X_{(2)} \geq cb_N) \geq 1 - \varepsilon$ for large N, and on that event $1 - \max_j w_j \leq \sum_{j \neq (1)} e^{X_j - X_{(1)}} \leq Ne^{-cb_N} \to 0$, because b_N grows polynomially. □

A.12 Proof of Proposition 4.7

$|s| \leq \kappa$ gives $Z(u) \in [e^{-\kappa}, e^{\kappa}]$, so $\mathrm{d}\nu^s_{v|u}/\mathrm{d}\mu_v = e^s/Z(u) \in [e^{-2\kappa}, e^{2\kappa}]$. Then $D_{\mathrm{KL}}\left(\nu^s_{v|u} \,\|\, \mu_v\right) = \mathbb{E}_\nu \log(\mathrm{d}\nu/\mathrm{d}\mu_v) \leq 2\kappa$ and $\mathbb{E}_{\mu_v}[(\mathrm{d}\nu/\mathrm{d}\mu_v)^2] = \mathbb{E}_\nu[\mathrm{d}\nu/\mathrm{d}\mu_v] \leq e^{2\kappa}$; the limit of ESS/$N$ follows from Theorem 4.5. □

Reproducibility. Tables 2 and 3 are produced by

`test_bed/theory/misspecification.py`,

and the experiments of Section 5.4 by

`test_bed/analysis/bank_sweep.py`.

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